SECTION A

TRUE & FALSE QUESTIONS (10 MARKS)

(**INSTRUCTION**: Write your answer either **TRUE (T)** or **FALSE (F)** based on the following statements.)

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1. Error detection and response include only internal hardware error under services provided by operating system. [ ]
2. Sharing system resources require the operating system to ensure that an incorrect program cannot have other programs to do incorrectly. [ ]
3. During program execution, only instructions must be loaded into main memory. [ ]
4. The Processor Manager of a modern operating system performs its tasks in isolation with other managers. [ ]
5. General-purpose operating system does not fulfill the requirement of hard real-time systems. [ ]
6. The File Manager is responsible for the creation of a process. [ ]
7. The overhead for context switching varies from machine to machine, depending on the processor size. [ ]
8. *fork*() system call is use to create a process under supervisor mode of the computer system. [ ]
9. A truth table diagram can tell what event causes a process to change states. [ ]
10. Operating system provides mechanism for process creation and process termination. [ ]

SECTION B

STRUCTURED QUESTIONS (50 MARKS)

(**INSTRUCTION**: Please answer all questions in the space provided)

QUESTION 1 [12 Marks]

a) Based on Figure 1, complete the diagram. [2 marks]

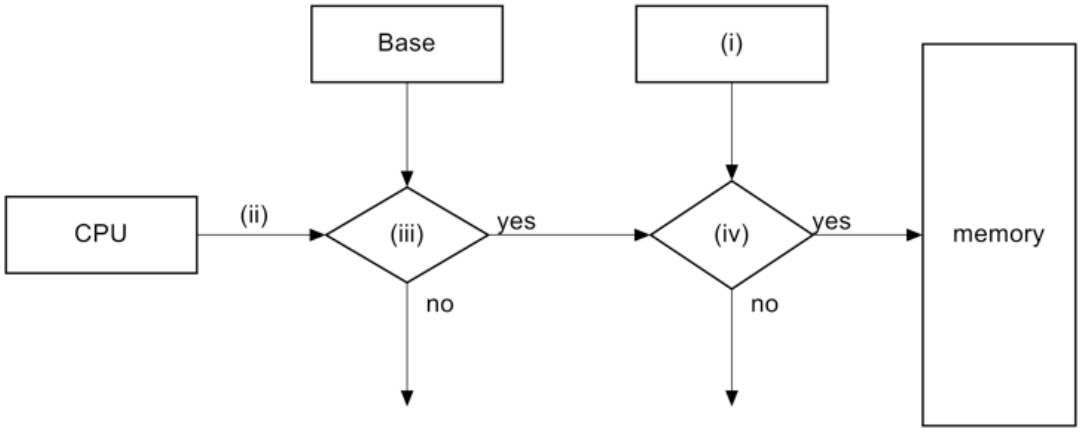


Figure 1: Memory Protection

Answer:

i.

ii.

iii.

iv.

b) Figure 2 shows four programs A, B, C and D, which program B and D have the same size of memory. Find the values and complete the table below for each registers for program B and D?

[4 marks]

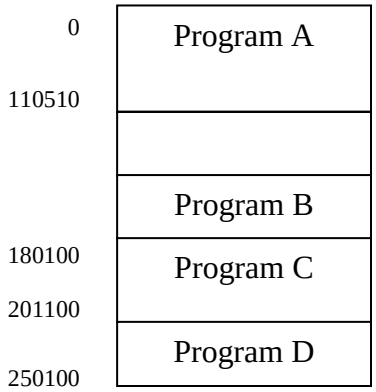


Figure 2

Answer :

Program	Base Register	Limit Register
B		
D		

c) Match hardware protection (either a / b / c / d) for descriptions in answer column. [6 marks]

Hardware Protection	Descriptions	Answer (either a / b / c / d)
a. Dual Mode **b. I/O Protection** **c. Memory Protection** **d. CPU Protection**	1. Base register hold the smallest legal physical address.	
	2. Instruction to switch interrupt services to on/off	
	3. Timer commonly used to implement time sharing.	
	4. User program should also be protected from other user programs.	
	5. Privileged instructions can be issued only in kernel mode	
	6. Time also used to compute the current time.	

QUESTION 2 [12 Marks]

a) List **FIVE** essential subsystem managers of modern operating systems. [5 marks]

Answer:

i.

ii.

iii.

iv.

v.

b) State the subsystem manager associated with the tasks of operating systems listed in the table below. [3 marks]

Task	Answer
Retains user access control	
Enforcing user/program resource access restrictions	
Dispatch a process from ready queue	

c) Match the list of terms provided with the descriptions in the table below. [4 marks]

Embedded System	UNIX	Supercomputer
Multiprogramming	Multitasking	Windows 8
Distributed System	Real-time System	VMware

Descriptions	Answer
An operating system used by mainframe systems	
Allows multiple operating systems to be used on a single hardware	
Allows multiple jobs to be executed concurrently with user interaction while a program is running	
Requires operating system with small kernel and flexible function capabilities	

QUESTION 3 [13 Marks]

a) Figure 3 shows a process in Five State Models.

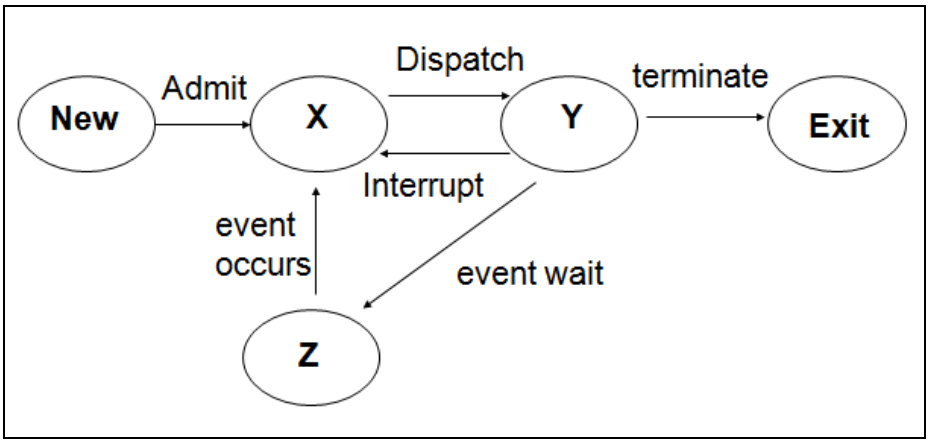


Figure 3

i) List state X, Y and Z. [1.5 marks]

Answer :

X :

Y :

Z :

ii) Explain state X and Z. [3 marks]

Answer :

b) List **THREE** categories of contents in Process Control Block (PCB). [1.5 marks]

Answer:

i.

ii.

iii.

c) Identify the correct commands in table below by selecting the answer options given. Assume you are in your working directory. [7 marks]

Answer Options:

A	`cat testfork1.c`
B	`gcc testfork1.c`
C	`mkdir Result`
D	`gcc testfork1.c –o testfork1result`
E	`ls`
F	`view testfork1.c testfork2.c`
G	`cp testfork1.c testfork2.c`
H	`a.out`
I	`cd Result`
J	`dir Result`

Tasks	Answer
Make another copy of testfork1.c file with name testfork2.c	
Compile testfork1.c and specify the output filename as testfork1result	
View the content of the file testfork1.c without using the editor.	

Create a directory with name Result	
List content of a directory	
Change working directory to Result	
What is default output filename generated after compiling the source code	

QUESTION 4 [13 Marks]

a) The following table shows the trace addresses for 3 processes, A, B and C and the dispatcher in memory. These processes are running concurrently on a single processor system. Process A request an I/O resource at instruction cycle 4 and wait up to 30 instruction cycles. Then, process B request an I/O resource at instruction cycle 35 and wait up to 20 instruction cycles.

The processes and location.

Process	Addresses
A	5000 – 5020
B	7000 – 7050
C	2000 – 2015
Dispatcher	200 – 204

Complete Figure 4 by drawing the timeline of the process state for all processes within 70 instruction cycles with THREE different indicators for **running**, **ready** and **blocked state**. Assume the CPU time = 6 instruction cycles. [6 marks]

Answer:

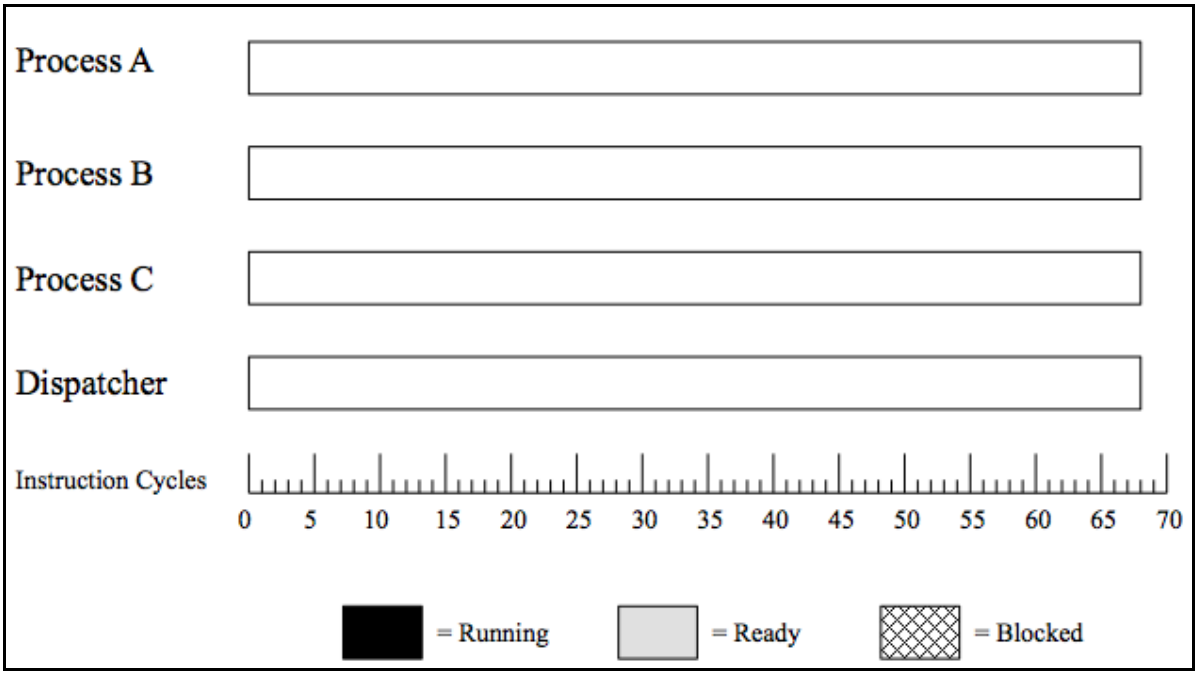


Figure 4

b) Consider program `processTree.c` in Figure 5. Draw the process tree for the execution of the program by drawing all processes created. Follow the example given in Figure 6 that must contain the value of new id of each new process and the expected output. Assume the initial process id = 3. [7 marks]

```
// processTree.c

     int main() {

     int pid1, pid2;

     pid1 = fork();
     print("Operating system");
     pid2 = fork();
     if (pid1 > pid2)
          print("is interesting");
     if (pid1 == pid2)
          print("I love OS");
     else
          print("Operating system");
```

Figure 5: A part of C programming language.

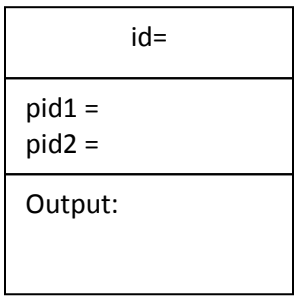


Figure 6: Example of each process.

Answer:

----------End of Question ----------